

LOGRONO AGONCILLO AP, Spain

WMO: 080840

Lat: 42.4522N

Lon: 2.3311W

Elev: 1191

StdP: 14.07

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	28.2	30.6	20.4	16.0	36.1	24.1	19.1	37.1	22.7	46.7	19.4	46.4	1.9	270	0.451

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	24.9	95.0	68.3	91.6	67.6	88.2	66.8	71.2	87.7	69.7	85.8	68.3	83.5	8.6	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.1	100.6	76.9	64.5	95.1	75.2	63.0	90.1	74.0	35.8	88.0	34.4	86.0	33.2	83.8	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
19.7	16.6	14.2	23.4	100.2	3.5	2.7	20.8	102.1	18.8	103.8	16.8	105.3	14.3	107.3
			22.6	73.9	3.6	2.6	20.0	75.8	17.9	77.3	15.9	78.8	13.3	80.7
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	43.9	45.4	50.8	54.8	61.3	69.4	73.6	73.3	67.1	59.5	49.6	43.9	(6)
(7)		DBStd	12.16	5.44	5.30	5.59	5.51	6.15	6.58	5.52	5.16	5.10	5.77	5.55	5.67	(7)
(8)		HDD50	698	199	145	59	18	2	0	0	0	4	73	198		(8)
(9)		HDD65	3478	654	550	441	309	148	31	5	3	34	187	463	653	(9)
(10)		CDD50	3535	10	15	83	161	351	582	731	723	513	297	60	9	(10)
(11)		CDD65	841	0	0	0	2	32	163	271	260	98	15	0	0	(11)
(12)		CDH74	8484	0	0	15	75	448	1761	2689	2496	868	131	1	0	(12)
(13)		CDH80	3630	0	0	1	9	123	783	1274	1152	273	15	0	0	(13)
(14)	Wind	WSAvg	5.6	6.1	6.4	6.6	6.2	5.8	5.5	5.3	5.1	4.7	4.8	5.7	5.7	(14)
(15)	Precipitation	PrecAvg	26.0	2.5	2.2	2.5	2.8	2.6	2.0	1.3	0.9	1.7	2.4	3.0	2.2	(15)
(16)		PrecMax	34.4	4.6	4.6	5.3	6.3	8.3	5.1	3.3	2.8	3.9	5.8	5.9	5.2	(16)
(17)		PrecMin	20.9	0.8	0.8	0.8	1.5	0.4	0.2	0.1	0.0	0.9	0.9	1.3	0.6	(17)
(18)		PrecStd	3.7	1.0	1.1	1.1	1.0	1.4	1.2	0.8	0.6	0.7	1.1	1.0	1.1	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	62.3	65.8	75.3	80.9	89.3	97.8	98.4	98.9	92.1	82.1	69.3	61.3	(19)
(20)		0.4%	MCWB	52.8	52.4	55.3	60.1	64.4	67.6	69.3	68.5	66.9	62.2	56.9	52.2	(20)
(21)		2%	DB	58.1	61.5	70.1	75.5	83.4	92.7	94.8	94.4	86.8	77.6	64.4	57.3	(21)
(22)		2%	MCWB	50.5	50.9	54.4	58.0	62.8	66.9	68.6	68.0	65.4	61.6	55.4	50.4	(22)
(23)		5%	DB	55.4	58.5	66.2	71.2	79.0	88.4	91.5	90.8	82.7	73.6	61.0	54.8	(23)
(24)		5%	MCWB	48.9	49.4	53.3	56.2	61.4	66.2	68.3	67.6	64.2	60.4	53.6	49.1	(24)
(25)		10%	DB	52.7	55.2	62.4	67.0	75.0	84.3	87.7	86.7	79.0	69.9	58.2	52.1	(25)
(26)		10%	MCWB	47.3	47.6	51.4	54.7	59.8	65.3	67.3	67.0	63.3	59.1	52.0	47.5	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	54.2	54.4	58.5	61.6	67.5	71.3	73.7	73.3	70.2	65.4	59.3	53.7	(27)
(28)		0.4%	MCDWB	60.3	61.8	69.9	77.0	84.3	88.4	91.9	89.9	83.4	76.5	66.0	58.6	(28)
(29)		2%	WB	51.6	52.0	55.8	59.3	64.6	69.3	71.4	71.1	68.0	63.6	56.6	51.7	(29)
(30)		2%	MCDWB	56.4	60.0	66.6	72.5	79.3	86.5	88.5	87.2	81.2	73.0	62.4	56.2	(30)
(31)		5%	WB	49.7	50.1	54.1	57.4	62.6	67.8	69.8	69.5	66.3	62.1	54.8	50.0	(31)
(32)		5%	MCDWB	54.5	57.0	64.0	68.7	76.0	84.0	86.2	85.0	78.4	70.2	59.5	53.6	(32)
(33)		10%	WB	47.9	48.3	52.2	55.4	60.8	66.2	68.2	68.0	64.6	60.6	53.0	48.3	(33)
(34)		10%	MCDWB	52.0	53.8	61.0	65.6	72.6	81.5	83.4	82.6	75.7	67.5	57.1	51.8	(34)
(35)	Mean Daily Temperature Range		MDBR	13.6	16.4	19.9	20.5	21.7	23.8	24.9	24.8	22.4	19.5	14.2	13.1	(35)
(36)		5% DB	MCDBR	18.3	22.9	27.3	29.1	30.1	31.4	31.4	31.7	28.8	25.4	18.7	16.8	(36)
(37)		5% DB	MCWBR	11.6	13.3	14.0	13.9	12.8	11.4	10.4	10.6	10.9	11.8	10.8	10.8	(37)
(38)		5% WB	MCDBR	16.3	20.6	24.6	25.7	26.9	28.2	28.0	27.2	24.8	20.9	16.3	14.9	(38)
(39)		5% WB	MCWBR	10.9	12.6	13.4	13.1	12.4	11.4	10.4	10.6	10.4	10.6	10.3	10.4	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.312	0.330	0.371	0.394	0.408	0.417	0.405	0.408	0.386	0.358	0.328	0.309	(40)	
(41)		taud	2.475	2.414	2.303	2.253	2.246	2.256	2.290	2.285	2.345	2.423	2.468	2.484	(41)	
(42)		Ebn at Noon	264	276	276	277	275	272	274	270	267	262	255	254	(42)	
(43)		Edh at Noon	23	29	37	42	43	43	41	40	35	29	23	22	(43)	
(44)	All-Sky Solar Radiation	RadAvg	522	780	1170	1501	1795	2070	2162	1880	1451	956	584	471	(44)	
(45)		RadStd	54	98	149	154	142	121	102	90	65	84	56	55	(45)	

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	-1.13	-1.51	N/A	N/A	N/A	N/A	(46)
(47)	Regional (10 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	(47)

Nomenclature: See separate page